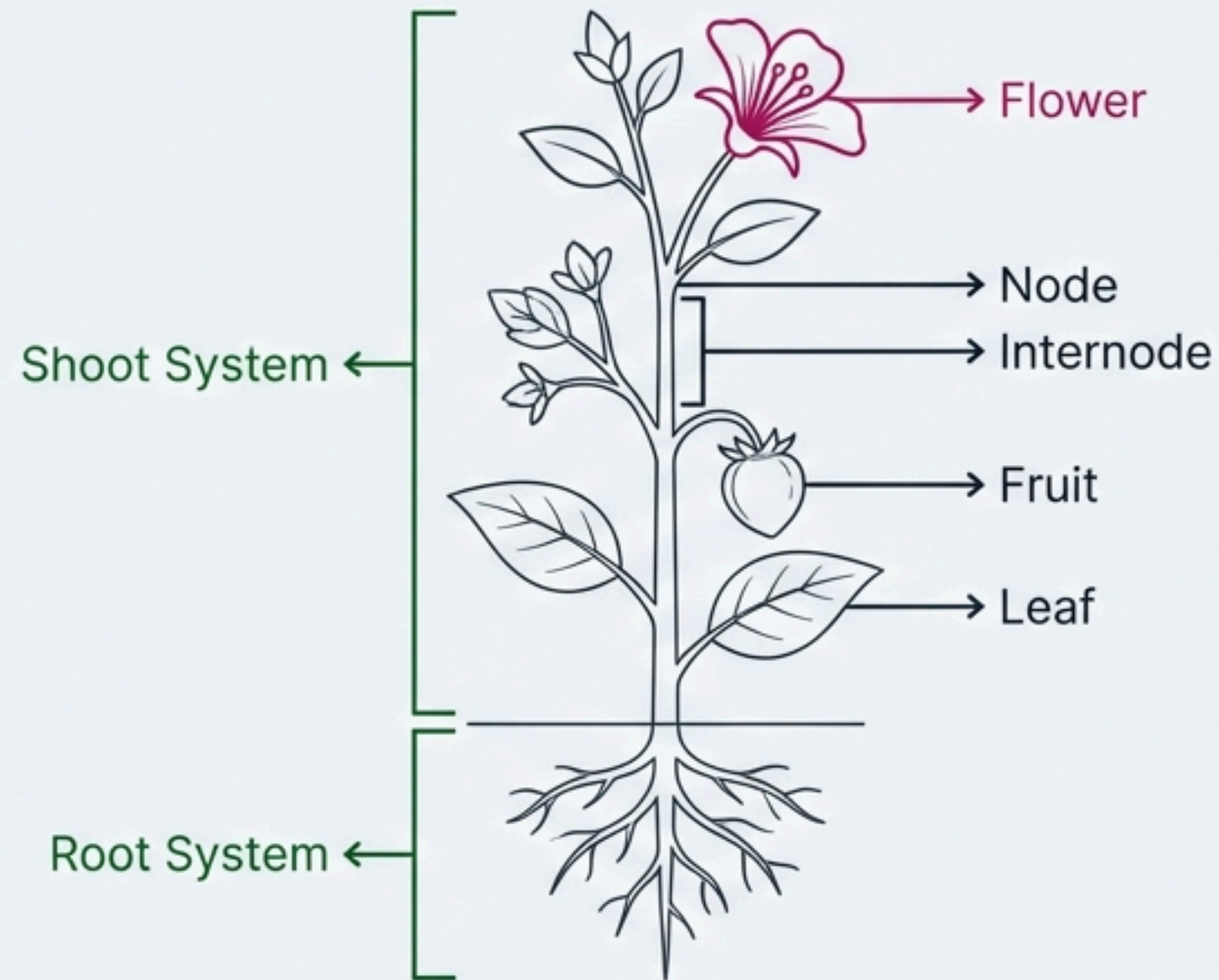




MORPHOLOGY OF **FLOWERING PLANTS**

MORPHOLOGY OF FLOWERING PLANTS

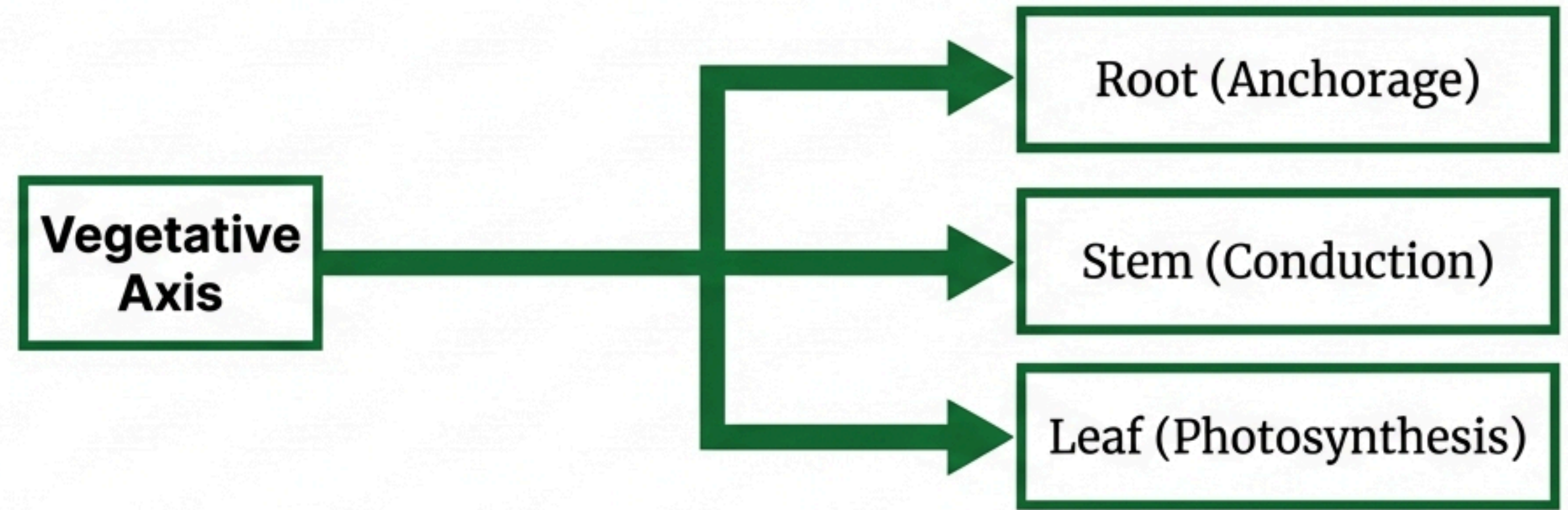
Complete NCERT Review for NEET-UG | Unit 2: Structural Organisation



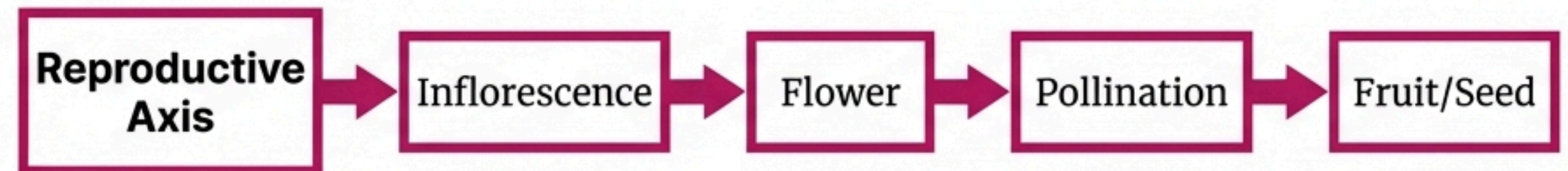
Chapter Overview & Syllabus Mapping

NCERT Scope

- 5.1 The Root
- 5.2 The Stem
- 5.3 The Leaf
- 5.4 The Inflorescence
- 5.5 The Flower
- 5.6 Fruit & Seed
- 5.9 Family Description (Solanaceae)



EXAM ALERT: NEET
Weightage: HIGH.
Expect 3-4 questions.
Focus on examples &
diagrams.

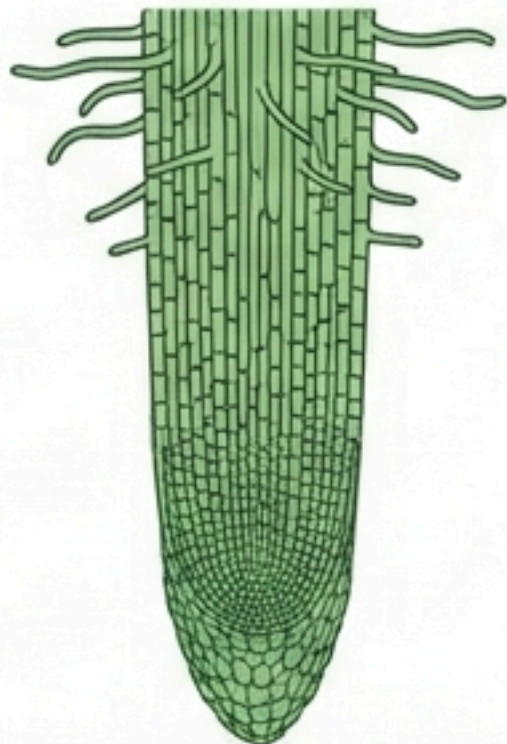


5.1 THE ROOT: Types & Regions

Types of Roots

Tap Root	Fibrous Root	Adventitious Root
• Direct elongation of primary root. • Example: Mustard (Dicot) 	• Primary root short-lived; replaced by roots from stem base. • Example: Wheat (Monocot) 	• Arise from parts other than radicle. • Examples: Grass, Monstera, Banyan

Regions of the Root Tip



Region of Maturation: Epidermal cells differentiate into Root Hairs (absorption).

Region of Elongation: Cells responsible for growth in length.

Region of Meristematic Activity: Small, thin-walled cells, dense protoplasm.

Root Cap: Thimble-like structure protecting apex.

5.2 STEM & 5.3 LEAF: Vegetative Morphology

Leaf Structure

- **Leaf Base:** Attachment point. Swollen 'Pulvinus' in Legumes; Sheath in Monocots.
- **Stipules:** Lateral small leaf-like structures.
- **Petiole:** Holds blade to light; flutters for cooling.



Venation Types

- **Reticulate (Dicots)**
Network of veins; characteristic of dicotyledonous plants.
- **Parallel (Monocots)**
Veins run parallel to each other; characteristic of monocotyledonous plants.

Phyllotaxy & Compound Leaves

- **Phyllotaxy (Arrangement):**
Alternate (China rose), Opposite (Guava), Whorled (Alstonia).
- **Palmately Compound:**
Leaflets at a common point (Silk Cotton).



5.4 INFLORESCENCE & FLOWER SYMMETRY

Inflorescence Types

Racemose

Main axis grows indefinitely. Flowers in **Acropetal** succession.



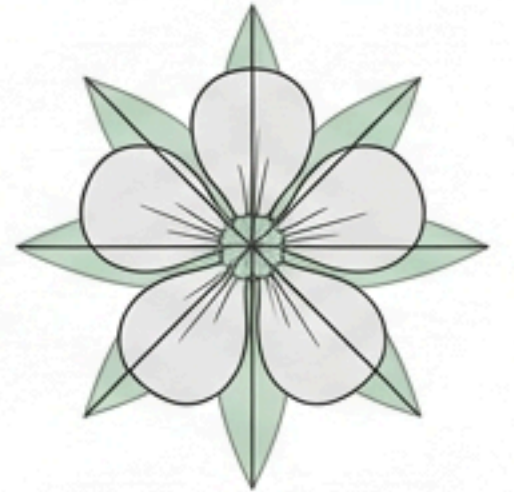
Cymose

Main axis terminates in flower (Limited).
Flowers in **Basipetal** order.

Flower Symmetry

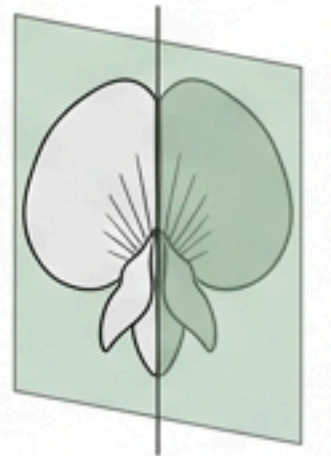
Actinomorphic (Radial)

Equal halves in any radial plane.
Mustard, **Datura**, **Chilli**.



Zygomorphic (Bilateral)

Similar halves in one vertical plane only.
Pea, **Gulmohur**, Bean.



Asymmetric

No symmetry.
Canna.



OVARY POSITION (Insertion on Thalamus)

Hypogynous

Superior Ovary

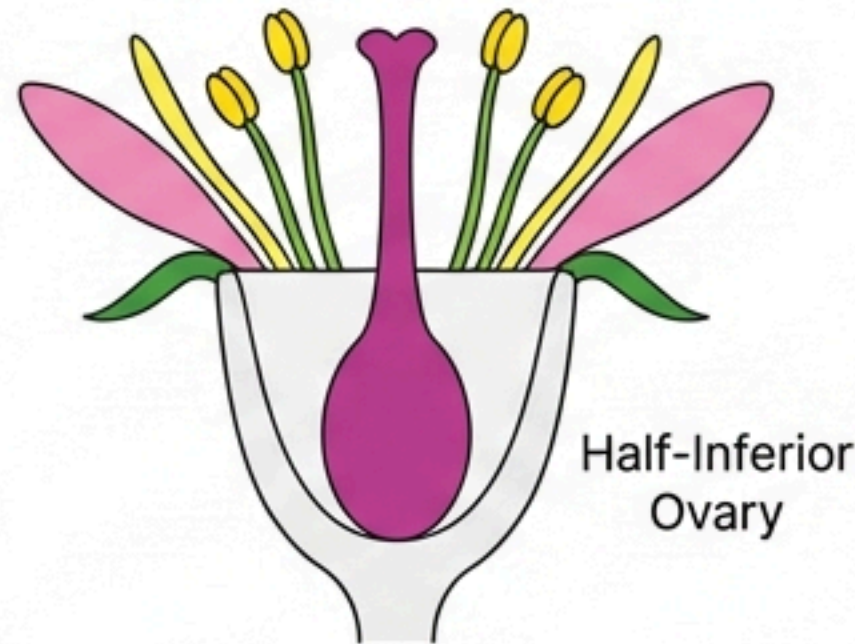


Gynoecium occupies highest position.

Examples: Mustard, China rose, Brinjal.

Perigynous

Half-Inferior Ovary



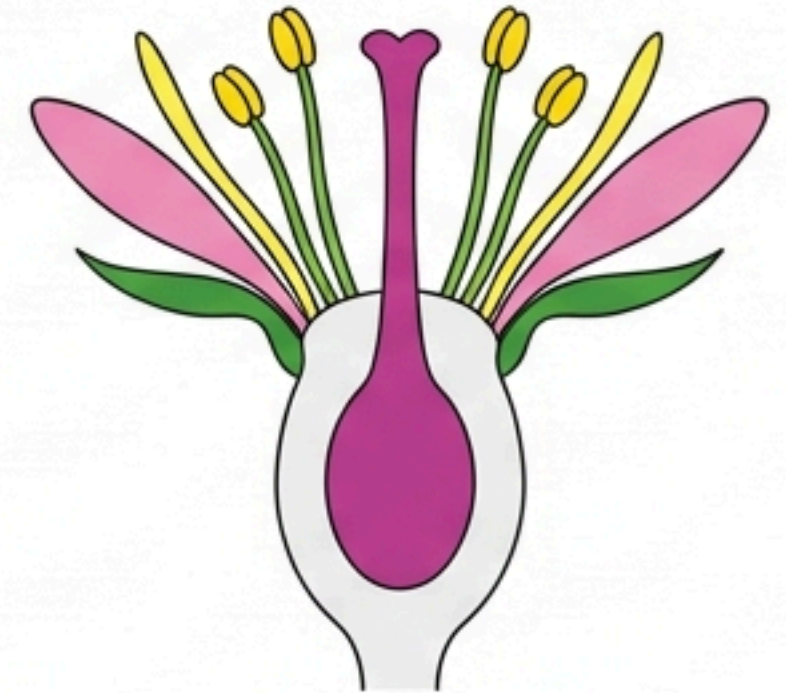
Parts on rim of thalamus at same level.

Examples: Plum, Rose, Peach.

Mnemonic: PPR

Epigynous

Inferior Ovary

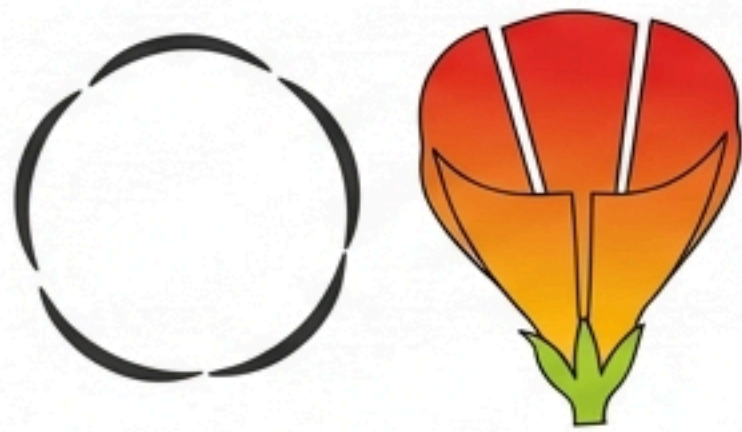


Thalamus fuses with ovary wall.

Examples: Guava, Cucumber, Sunflower ray florets.

5.5.1 FLORAL PARTS: Aestivation & Androecium

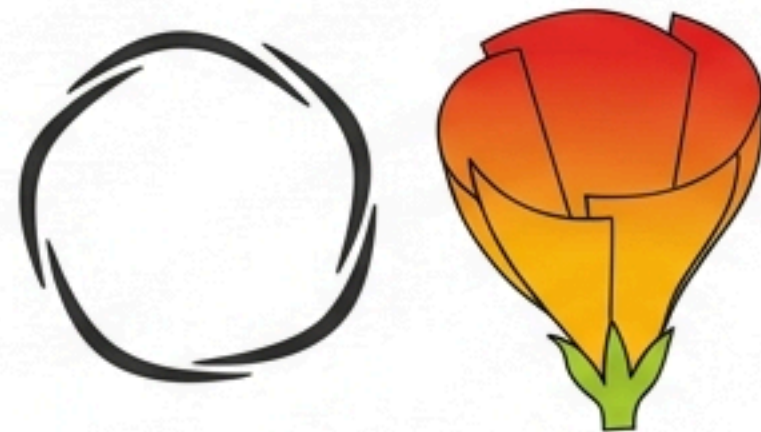
Aestivation (Sepal/Petal Arrangement)



Valvate

(Margins touch)

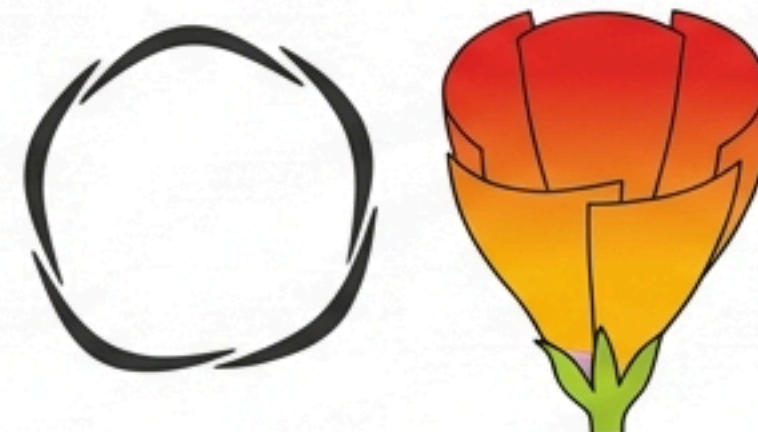
Example: **Calotropis**



Twisted

(Regular overlap)

Example: **China rose**



Imbricate

(Irregular overlap)

Example: **Gulmohur**

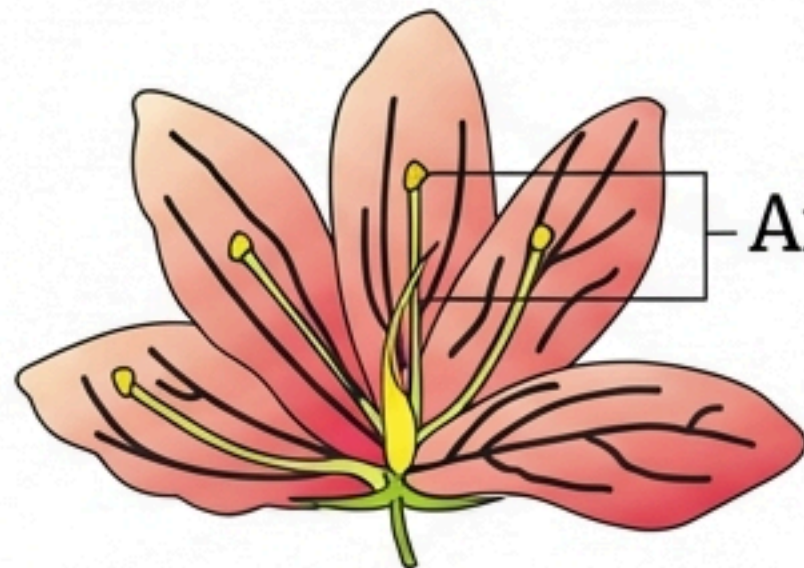


Vexillary

(Standard, Wings, Keel)

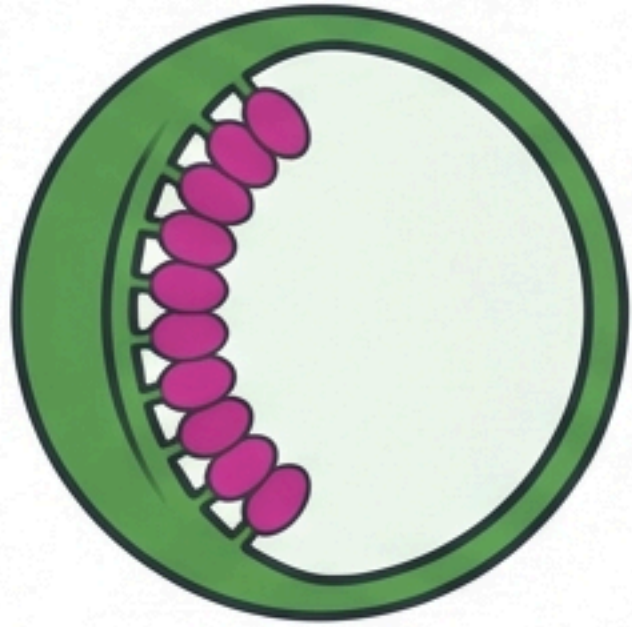
Example: **Pea**

Androecium (Stamens)



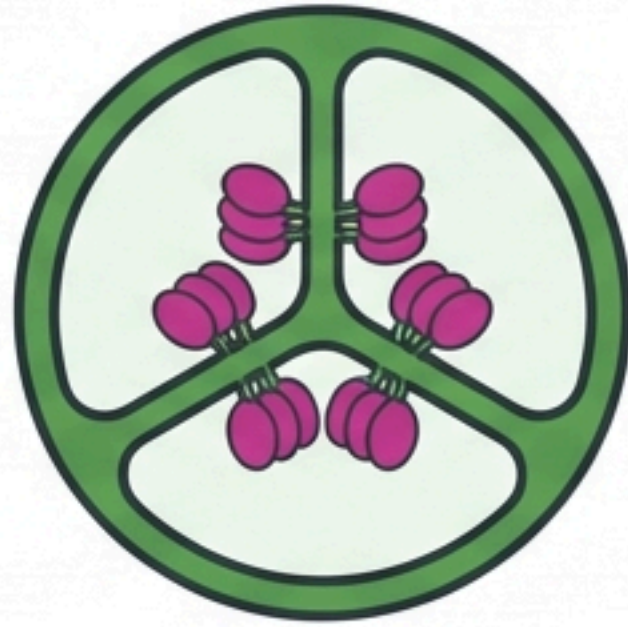
- Adhesion: **Epipetalous** (**Brinjal**), **Epiphyllous** (**Lily**).
- Cohesion (Bundles):
 - **Monadelphous** (**China rose**)
 - **Diadelphous** (**Pea**)
 - **Polyadelphous** (**Citrus**)

5.5.1.4 PLACENTATION (Arrangement of Ovules)



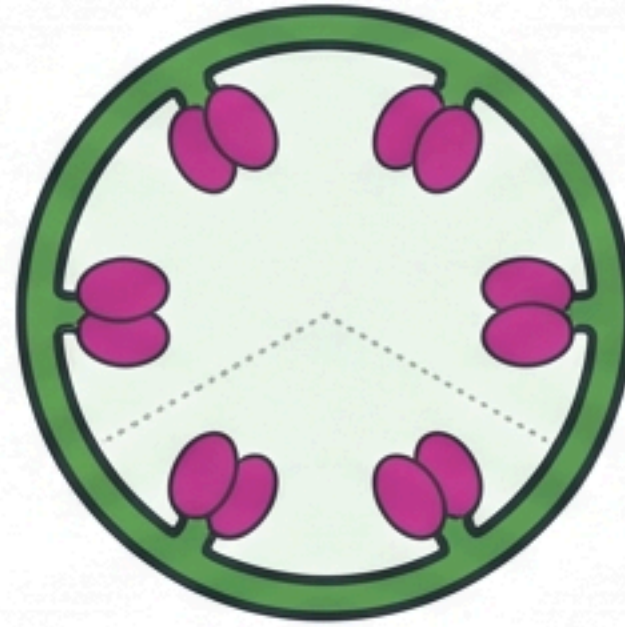
Marginal

Ridge along ventral suture.
Example: Pea.



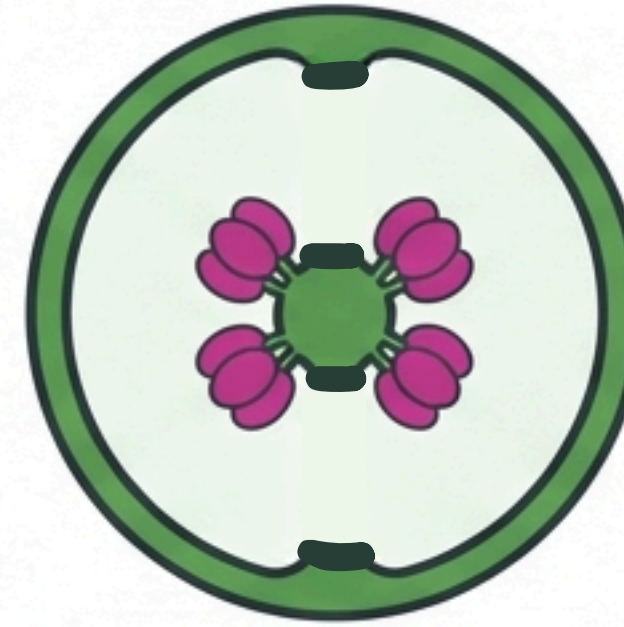
Axile

Multilocular, central axis.
Examples: China rose, Tomato, Lemon.



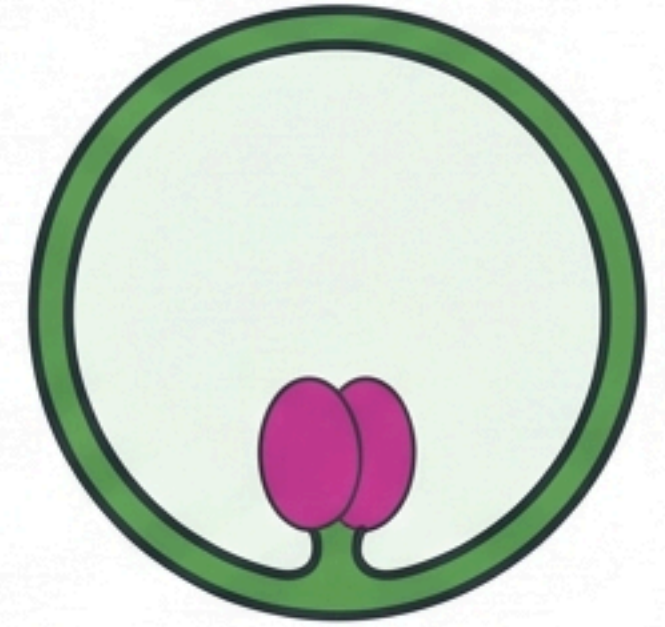
Parietal

Inner wall, false septum.
Examples: Mustard, Argemone.



Free Central

Central axis, no septa.
Examples: Dianthus, Primrose.

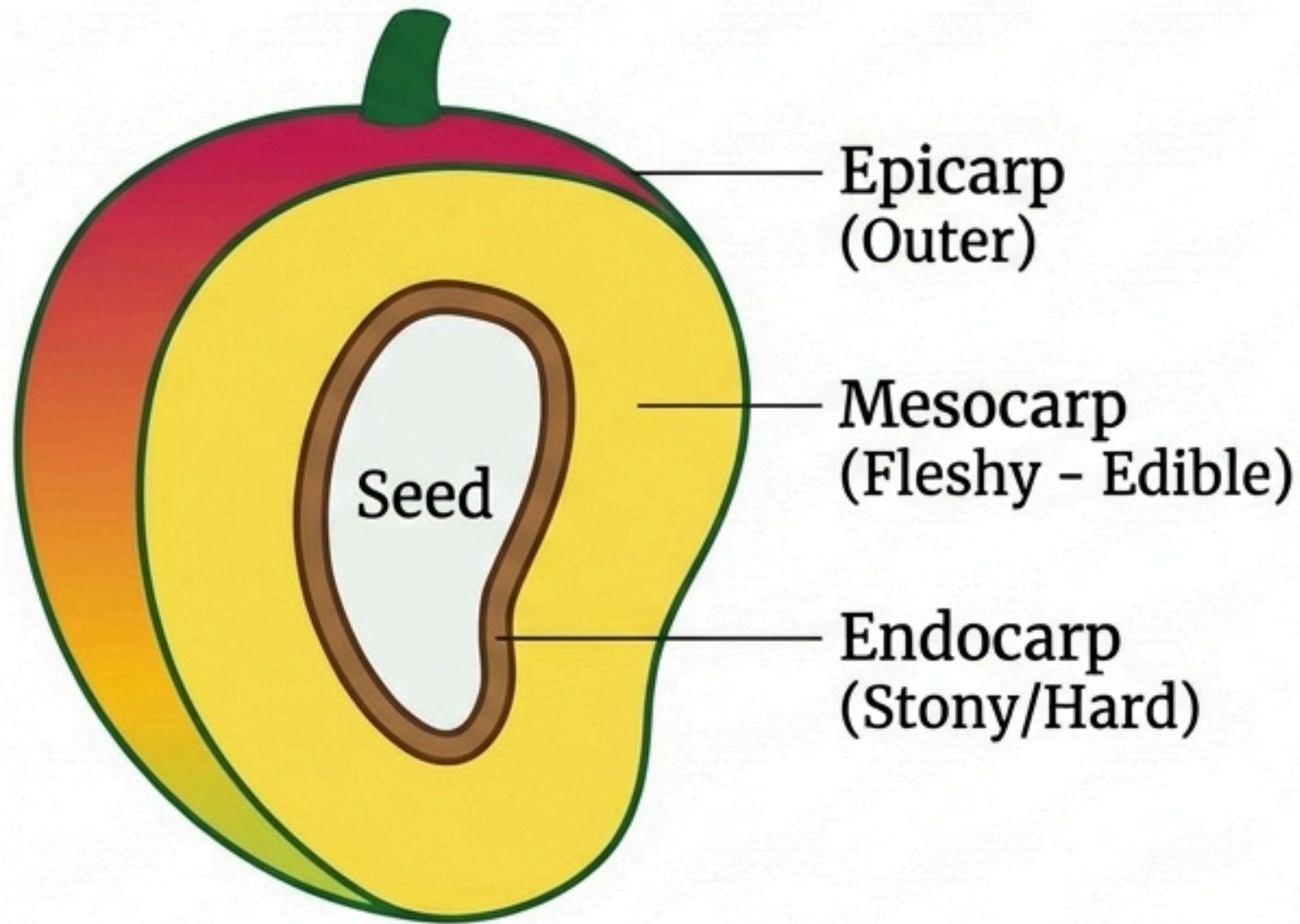


Basal

Base of ovary, single ovule.
Examples: Sunflower, Marigold.

5.6 FRUIT & 5.7 SEED

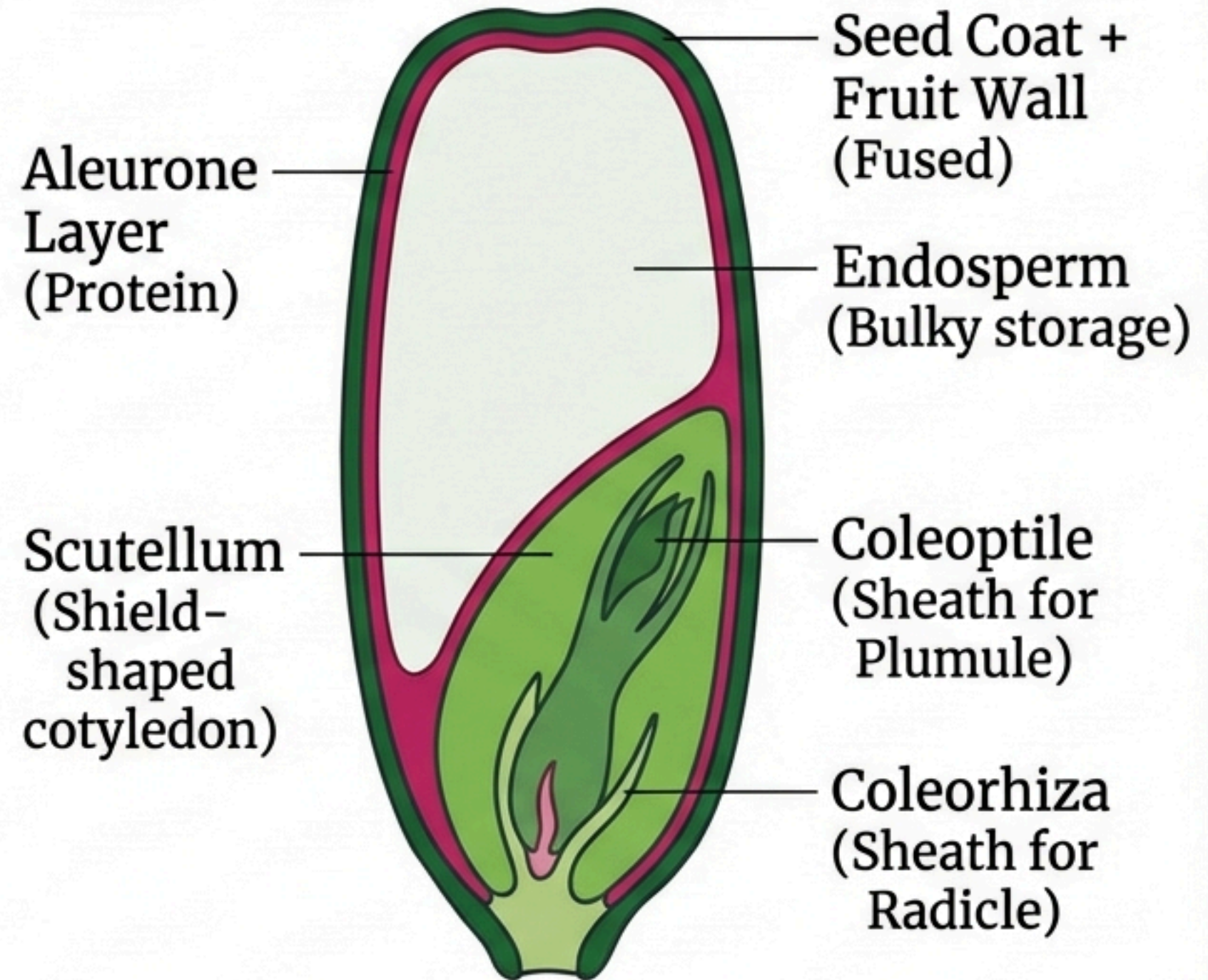
The Drupe Fruit



Developed from monocarpellary superior ovary.

Coconut: Mesocarp is fibrous.

Monocot Seed Structure (Maize)



5.9 FAMILY: SOLANACEAE ('Potato Family')

Floral Formula: $\oplus \text{♀} K_{(5)} C_{(5)} A_5 G_{(2)}$

Column 1 Vegetative Characteristics	Column 2 Floral Characteristics	Column 3 Economic Importance
• Stem: Herbaceous, often underground (Potato). • Leaves: Alternate, Reticulate venation.	• Inflorescence: Cymose (Solanum). • Flower: Bisexual, Actinomorphic. • Calyx/Corolla: 5 united, Valvate. • Gynoecium: Bicarpellary, Syncarpous, Superior. • Placenta swollen with many ovules. Oblique septum.	• Food: Potato, Tomato, Brinjal. • Spice: Chilli. • Medicine: Belladonna, Ashwagandha. • Ornamental: Petunia. • Fumigatory: Tobacco.

⚠️ CRITICAL NCERT LINES (Direct NEET Pickups)

- “ The root is covered at the apex by a ‘**thimble-like structure**’ called the root cap. ”
- “ In monocotyledons, the leaf base expands into a ‘**sheath covering the stem partially or wholly**’. ”
- “ Long thin flexible petioles allow leaf blades to ‘**flutter in wind**’, thereby cooling the leaf. ”
- “ Ovary is one-chambered but it becomes two-chambered due to the ‘**formation of the false septum**’ (e.g., Mustard). ”
- “ In coconut which is also a drupe, the ‘**mesocarp is fibrous**’. ”

CONCEPTUAL COMPARISONS & DIFFERENCES

Racemose vs. Cymose

Feature	Racemose Inflorescence	Cymose Inflorescence
Growth	Unlimited	Limited
Order	Acropetal	Basipetal

Apocarpous vs. Syncarpous

Feature	Apocarpous Gynoecium	Syncarpous Gynoecium
Condition	Free Carpels	Fused Carpels
Examples	Lotus, Rose	Mustard, Tomato

Stem Hair vs. Tyne-Hair

Feature	Apocarpous Gynoecium	Root Hair Gynoecium
Origin	Exogenous	Endogenous
Cellularity	Multicellular	Unicellular

Stem Hair vs. Root Hair

Feature	Stem Hair (Trichomes)	Root Hair
Origin	Exogenous	Endogenous
Cellularity	Multicellular	Unicellular

HIGH YIELD EXAMPLES & EXAM TRAPS

Confusing Pairs

- **Inter Epipetalous** (**Stamen on Petal**): **Brinjal**
- **Inter Epiphyllous** (**Stamen on Perianth**): **Lily**

The 'Superior' Trap

Superior Ovary = Hypogynous Flower.
Do not confuse 'Superior' with the flower type.

Memory Hooks

- **Free Central**: 'Free DP' -> **Dianthus, Primrose.**
- **Basal Placentation**: 'Basal is Best (Sun)' -> **Sunflower, Marigold.**
- **Perigynous**: 'P-P-P-R' -> **Plum, Peach, Rose.**
- **Whorled Phyllotaxy**: 'World of **Alstonia**'.

STATEMENT ANALYSIS & ASSERTION-REASON

Statement 1

"Leaves originate from shoot apical meristems and are arranged in an acropetal order."

TRUE.

(Note: Often confused with basipetal order of cymose flowers).

Assertion-Reason

Assertion: The ovary in Guava is inferior.

Reason: The thalamus grows upward enclosing the ovary completely and fuses with it.

Both True. Reason is the Correct Explanation for Assertion (Epigynous).

Statement 2

"In Maize, the seed coat is membranous and fused with the fruit wall."

TRUE.

(Characteristic feature of Caryopsis fruit).

FINAL CHAPTER SNAPSHOT: The 'Must-Remembers'

Mustard (Brassicaceae)

- Hypogynous (Superior)
- Syncarpous
- Parietal Placentation (False Septum)
- Tetramerous

China Rose (Malvaceae)

- Twisted Aestivation
- Monoadelphous Stamens
- Axile Placentation
- Alternate Phyllotaxy

Pea (Fabaceae)

- Zygomorphic
- Vexillary Aestivation
- Diadelphous Stamens
- Marginal Placentation

Solanaceae

- Epipetalous
- Swollen Placenta
- Oblique Ovary
- Axile Placentation

Success in Morphology lies in memorizing the Example Sets.